Niklas Erdmann



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linkedin, google scholar, homepage, orcid, github

About Me: Personal curiosity, peers, and experiences during my prior education sparked my interest in applying Artificial Intelligence (AI) to our world's problems and brought me to pursue a PhD in Oslo. Since then, I have enjoyed many cross-disciplinary collaborations, in which I contribute my broad background in AI, Statistics, and Psychology. A continuous passion for research and teaching draws me towards more academic endeavours.

Post-Graduate Education:

PhD, Dep. of Technology Systems, University of Oslo **03.2023 - 04.2027**
Title: *Artificial Intelligence-based Prognosis of Dynamic Energy Systems*

A full-time salaried position (4-year contract incl. 1 year of teaching) applying machine learning to energy forecasting for photovoltaic (PV) systems. It focuses on probabilistic forecasting, with a further interest in monotonic neural networks and their applications, in collaboration with the Institute for Energy Technology (IFE) and the Norwegian Defence Research Establishment (FFI).

Publications

Erdmann, N., Bentsen, L., Stenbro, R., Riise, H. N., Warakagoda, N. D., & Engelstad, P. E. (2026). Multi-Horizon Time Series Forecasting of Non-Parametric CDFs with Deep Lattice Networks. In *Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence.* (arXiv, Code and Data)

Eriksen, E. W., Nygård, M. M., Erdmann, N., & Riise, H. N. (2026). Deep Learning Multihorizon Irradiance Nowcasting: A Comparative Evaluation of Three Methods for Leveraging Sky Images. *IEEE Journal of Photovoltaics*, *16*(4), 594–604. (DOI)

Erdmann, N., Bentsen, L. Ø., Stenbro, R., Riise, H. N., Warakagoda, N., & Engelstad, P. (2024). Deep and Probabilistic Solar Irradiance Forecast at the Arctic Circle. In *Proceedings of the 2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC)* (pp. 1599–1606). (IEEE, arXiv; Best Poster Award)

Fordham, D., & Erdmann, N. (2026). Estimating Lithium-Ion Battery State-of-Health with Monotonic Lattice Networks. *SSRN preprint*. (DOI)

Zieglmeier, S., Erdmann, N., & Warakagoda, N. D. (2025). Reinforcement Learning for Pollution Detection in a Randomized, Sparse and Nonstationary Environment with an Autonomous Underwater Vehicle. *arXiv preprint arXiv:2510.26347*. (arXiv)

Education:

MSc. Artificial Intelligence, University of Groningen **09.2019 - 07.2022**
English-taught, Grade: 8.4/10, ECTS: 121
Master Thesis: *"Dynamic Modeling of a P(VDF-TrFE-CTFE)-Composite Soft Actuator"* (RUG): dynamic modelling of an electro-active actuator from noisy experimental time series with echo state networks.

Completed courses across the AI spectrum: Data Science, Machine/Deep Learning, Computer Vision, Pattern Recognition, Multi-Agent Systems, Reinforcement Learning, and Neuroscience.

Publication

D'Anniballe, R., Erdmann, N., Selleri, G., & Carloni, R. (2022). Dynamic Modeling of P (VDF-TrFE-CTFE)-based Soft Actuators via Echo State Networks. In *Proceedings of the 2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (pp. 118–124). (IEEE)

BSc. Psychology, University of Groningen **09.2015 - 01.2019**
English-taught, Grade: 7.6/10, ECTS: 200

Excellence track of the Honours Programme, with a focus on Statistics, Research Methods, and Cognitive Psychology, and a minor in Artificial Intelligence. Followed by a Pre-Master in Artificial Intelligence (09.2018 - 08.2019, Grade 8.1/10).

Teaching Experience:

PhD Teaching Component, University of Oslo **09.2024 - 04.2027**
Lectures, assignment design and exams for TEK5040 Deep Learning for Autonomous Systems. Independent supervision of 5 Master's thesis projects (3 completed, 2 ongoing).

Teaching Assistant, University of Groningen **09.2019 - 07.2021**
Led tutorial sessions, graded assignments and exams, and coordinated a group of other TAs.

Teaching Qualifications:

Basic Competence in University Pedagogy, UiO **expected 04.2027**
Foundation module including peer mentoring (120 h) and the Research Supervision elective (30 h); the formal teaching qualification for academic staff at Nordic universities.

Teaching and Learning in STEM, UiO **2025**
30 h course on constructive alignment, backward design and the scholarship of teaching and learning.

Other Activities:

Talks and Posters

Poster (on site) and online talk, 40th AAAI Conference on Artificial Intelligence, Singapore (2026).
Invited talk, "Forecasting Uncertainty with ML", FME Solar Energy (2025).

Awards

Best Poster Award, IEEE 52nd Photovoltaic Specialist Conference (PVSC) (2024).
Best Paper Award, PhD educational component, University of Oslo, for the AUV pollution-detection paper.
Best Performer: Leukemia Challenge, biomarker classification, MSc AI, University of Groningen.

Academic Service

PhD/PostDoc Representative, Department Board, Dep. of Technology Systems (2025); vice-representative, Faculty Board, Faculty of Mathematics and Natural Sciences (2024), University of Oslo.

Relevant Skills and Experiences:

Languages
German (Native), English (Professional; fully English-taught education since 2015), Dutch & Norwegian (Elementary).

Programming
Python (primary) with PyTorch, TensorFlow, SciPy, Optuna and OpenCV.
Also: C, R, Matlab; Git, LaTeX, Slurm (HPC).
Agentic coding for research software (Claude Code).